

TECHNICAL REFERENCE

Zephyr Hood Remote

315 MHz RF Protocol / Reference and Implementation Notes

RESTRICTED

FORM ZH-RF-01
First Edition
October 2026
Home Automation
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SUMMARY

The four-button remote supplied with the Zephyr Lux Glo Island hood (model DLG-E36AWX) transmits a 24-bit fixed code by on-off keying at 315 MHz. Each word is a 16-bit address, 0x0514, followed by an 8-bit command, sent with a pulse-width code whose unit is 270 microseconds and closed by a sync pulse 31 units long. The remote repeats the word seven or eight times per press. There is no checksum, counter, rolling code or return channel, so the hood can be operated by replaying the four codes, and its state cannot be read over this link. An ESP32 with a CC1101 transceiver running ESPHome both decodes the remote and transmits the codes; on the ESP32 the shared data pin must be driven open-drain for transmission to work.

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1 Summary

The remote is a one-way, fixed-code transmitter of the PT2262 and EV1527 family. Every press of a given button produces the same 24-bit word, and the hood keeps all of its own state. The protocol was established on 27 October 2026 by receiving the remote with a CC1101 and decoding 709 bits from four presses; the transmit path was then confirmed by operating the hood from Home Assistant.[1]

1 ESP32 DevKit V1 with an Ebyte E07-M1101D CC1101 module, ESPHome 2026.10.0.

PROTOCOL AT A GLANCE

RADIO

Carrier	315 MHz
Modulation	OOK
Direction	One way, remote to hood

LINE CODE

Unit T	270 us (measured 269.6)
Bit 0	1T on, 3T off
Bit 1	3T on, 1T off
Sync	1T on, 31T off

FRAMING

Word	24 bits, MSB first, then sync
Word length	128T = 34.6 ms
Repeats per press	7 to 8
Address	0x0514
Integrity	None

WARNING

Fixed code. Anyone within radio range who knows a code can operate the hood. Treat the codes in this document like a garage-door code.

2 Command Set

Four buttons each send one code. The light and the fan step through their levels inside the hood, so a press means “next level”, never “set level”.

Table 1. Command codes for remote address 0x0514.

BUTTON	CODE	COMMAND BYTE	HOOD ACTION
Light	0x051430	0011 0000	Light: off, low, medium, high, off
Fan	0x051403	0000 0011	Fan: off, low, medium, high, off
Power	0x05140C	0000 1100	Fan and light off
Delay off	0x0514C0	1100 0000	Starts the delay-off timer

Pressing Fan and Delay off together puts the hood into Wi-Fi pairing mode, according to Zephyr’s Connect guide for DLG models. If the two digits combine as on other PT2262 remotes, that code is 0x0514C3. It has not been captured.

3 Physical Layer

The remote runs from a 23A 12-volt cell and carries the FCC grantee code VVP, which belongs to Leyant Industry, Zephyr's manufacturer. A 433.92 MHz filing exists for Zephyr part 14000010 (FCC ID VVP-14000010), but it describes a different remote: nothing from this one was received at 433.92 MHz in any modulation tried.

At 315 MHz the band was silent except for the remote. Four presses, one second apart, produced four isolated bursts of 110 to 150 pulses each, read on an edge counter at one-second resolution.[2]

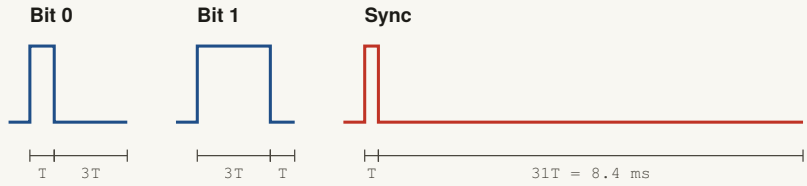
2 Two unrelated 433.92 MHz sensors nearby transmit every 30 and 50 seconds, with 425 and 2150 microsecond pulses. They were briefly mistaken for the remote.

Table 2. Receiver settings used for capture.

SETTING	VALUE
Frequency	315.000 MHz
Modulation	ASK/OOK
Filter bandwidth	812 kHz
Receiver glitch filter	150 us
End-of-frame idle	25 ms

4 Line Code

Every bit occupies four units. A zero is a short mark and a long space; a one is a long mark and a short space. Each word ends with a sync: one unit of carrier, then thirty-one units of silence.



Measured on 709 bits: short mark 270 ± 14 µs, long mark 815 ± 10 µs, sync gap 8383 ± 8 µs. Carrier on = high.

Figure 1. Line code. One unit T is 270 microseconds; carrier on is drawn high.

Table 3. Measured element durations over 709 bits and 27 sync pulses, in microseconds.

ELEMENT	NOMINAL	MEAN	STD. DEV.	MIN	MAX
Short mark	270	270	14	220	300
Long mark	810	815	10	790	840
Short space	270	265	10	240	290
Long space	810	806	17	750	940
Sync mark	270	271	12	260	290
Sync gap	8370	8383	8	8370	8400

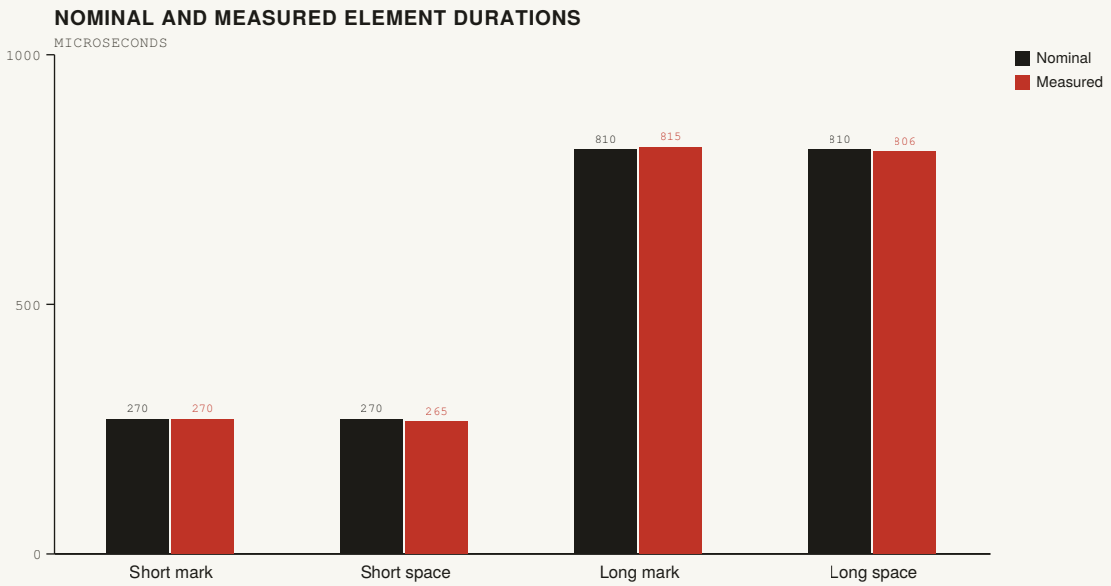


Figure 2. Measured values sit within 2 percent of the 270 microsecond unit.

The decoder in this repository accepts a short element of 120 to 480 microseconds, a long element of 560 to 1100, and a sync gap of five milliseconds or more. It decodes words with timing errors of plus or minus 20 percent.

5 Word Format

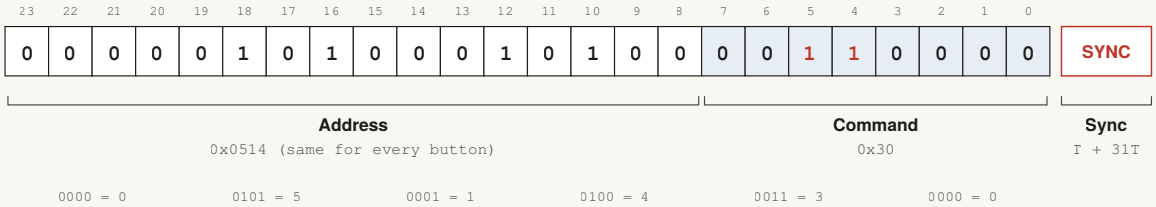


Figure 3. Word format, shown for the Light code 0x051430. Command bits set to one are in red.

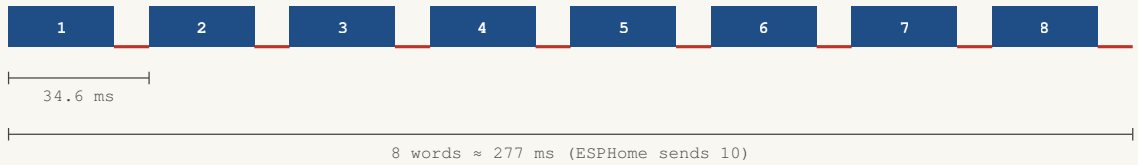
The layout is that of the PT2262 encoder, in which each tri-state digit is sent as two bits: 00 for zero, 11 for one, and 01 for floating.

Table 4. The codes read as PT2262 tri-state digits.

FIELD	BITS	TRI-STATE DIGITS
Address 0x0514	00 00 01 01 00 01 01 00	0 0 F F 0 F F 0
Light 0x30	00 11 00 00	0 1 0 0
Power 0x0C	00 00 11 00	0 0 1 0
Fan 0x03	00 00 00 11	0 0 0 1
Delay off 0xC0	11 00 00 00	1 0 0 0

Each button sets exactly one data digit. Eight tri-state address digits allow 6,561 addresses; the hood stores the one it accepts when a remote is paired to it.

6 Press Timing



Blue: 24 data bits (25.9 ms). Red: sync gap (8.4 ms). There is no preamble, checksum, counter or rolling code.

Figure 4. One press: the same 34.6 ms word sent back to back. Blue is data, red is the sync gap.

A tap sends seven or eight identical words, about a quarter of a second of transmission. The hood needs one clean word. The ESPHome transmitter sends ten, which the hood treats as a single press.

7 State and Feedback

The protocol has no status request and no reply. The hood has a receiver and no transmitter on this band, and the remote has no receiver. Hood state exists only inside the hood and, for hoods joined to the Zephyr Connect app, in Zephyr's cloud service, which keeps it in an AWS IoT device shadow reached over TLS with a per-device certificate. No local network interface to that state is documented.

The Home Assistant integration in this repository therefore behaves as a second remote: each button sends one press, and nothing is tracked.

8 Implementation

TRANSCEIVER CONFIGURATION

HARDWARE

Controller	ESP32 DevKit V1 (WROOM-32)
Radio	Ebyte E07-M1101D, CC1101
Antenna	SMA whip, 433 MHz

RADIO

Frequency	315 MHz
Modulation	ASK/OOK
Output power	+10 dBm
RX bandwidth	812 kHz

DATA PATH

Pin	GDO0 to GPIO4, shared TX and RX
Pin mode	Open drain, pull-up
Words per press	10

IMPORTANT

Open drain on the ESP32. With the cc1101 gdo0_pin option sharing GPIO4 (automatic mode switching), reception worked but the hood ignored every transmission. Configuring the shared pin as open drain with a pull-up, as the ESPHome CC1101 documentation requires on the ESP32, made the first transmission work.

The 433 MHz module and antenna operated the hood at 315 MHz from directly beneath it. If range from the final mounting position is marginal, a 315 MHz module and antenna is a direct replacement with no change to the configuration.

9 Method

The protocol was found by elimination. A published project for an older Zephyr remote pointed to 433.92 MHz, which proved wrong; a frequency sweep driven from Home Assistant located the remote at 315 MHz, and raw captures relayed through a Home Assistant sensor were decoded off-line.

PATH TO A WORKING TRANSMITTER

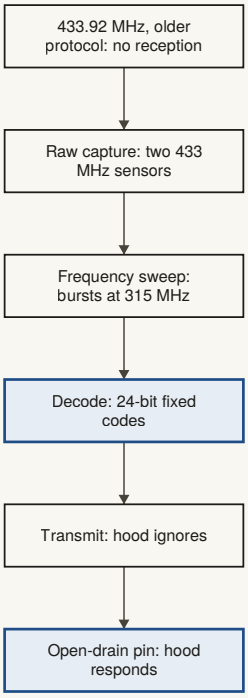


Figure 5. Each step was confirmed against Home Assistant history before the next.

Table 5. Sources.

SOURCE	USE
ESPHome CC1101 component documentation	Radio driver and ESP32 wiring method
HomeSpan ZephyrVentHood project	Older 433 MHz Zephyr protocol, ruled out
Zephyr Connect quick start, 2025	Pairing combination for DLG models
FCC ID VVP-14000010	433.92 MHz Zephyr remote, a different model